



LAB 4: STIMULATING RADIOACTIVE DECAY PROCESS

PERSONAL AND GROUP INFORMATION REDACTED

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```
clc; clear all;
```

Base Simulation (Given Code)

```
n      = 100;
N0     = 1000;
tau    = 1.0;
Delta  = 0.02;

N      = zeros(1, n+2);
time   = zeros(1, n+2);
N(1)   = N0;
time(1) = 0;

for i = 1:n+1
    N(i+1) = N(i) - Delta * N(i) / tau;
    time(i+1) = i * Delta;
end
```

Part (i): Half-Life Determination

```
half_N = N0 / 2;

idx_half = find(N <= half_N, 1, 'first');
t1 = time(idx_half - 1); t2 = time(idx_half);
N1 = N(idx_half - 1); N2 = N(idx_half);
t_half_exp = t1 + (half_N - N1) * (t2 - t1) / (N2 - N1);

t_half_theory = tau * log(2);
N_at_half_theory = N0 / 2;

disp('--- Half-Life Results ---')
fprintf('Theoretical : t_half = %.4f s | N(t_half) = %.2f atoms\n', ...
        t_half_theory, N_at_half_theory);
fprintf('Experimental : t_half = %.4f s | N(t_half) = %.2f atoms\n', ...
        t_half_exp, half_N);
fprintf('Relative error = %.4f%%\n\n', ...
        abs(t_half_exp - t_half_theory)/t_half_theory * 100);

--- Half-Life Results ---
Theoretical : t_half = 0.6931 s | N(t_half) = 500.00 atoms

Experimental : t_half = 0.6862 s | N(t_half) = 500.00 atoms
Relative error = 0.9971%
```

RESULTS FOR PART (i)

Theoretical:

(a) *half – life* = 0.6931 s

(b) *No of atoms remaining* = 500.00 atoms

Experimental:

(a) *half – life* = 0.6862 s

(b) *No of atoms remaining* = 500.00 atoms

Relative error = 0.9971%

Part (ii): Time Constant Determination

```
target_N = N0 / exp(1);

idx_tau = find(N <= target_N, 1, 'first');
t1 = time(idx_tau - 1); t2 = time(idx_tau);
N1 = N(idx_tau - 1); N2 = N(idx_tau);
tau_exp = t1 + (target_N - N1) * (t2 - t1) / (N2 - N1);

tau_theory = tau;
N_at_tau_theory = N0 / exp(1);

disp('--- Time Constant Results ---')
fprintf('Theoretical : tau = %.4f s | N(tau) = %.2f atoms\n', ...
        tau_theory, N_at_tau_theory);
fprintf('Experimental : tau = %.4f s | N(tau) = %.2f atoms\n', ...
        tau_exp, target_N);
fprintf('Relative error = %.4f%%\n\n', ...
        abs(tau_exp - tau_theory)/tau_theory * 100);

--- Time Constant Results ---
Theoretical : tau = 1.0000 s | N(tau) = 367.88 atoms
Experimental : tau = 0.9900 s | N(tau) = 367.88 atoms
Relative error = 0.9983%
```

RESULTS FOR PART (ii)

Theoretical:

(a) Time constant = 1.0000 s (b) No of atoms remaining = 367.88 atoms

Experimental:

(a) Time constant = 0.9900 s (b) No of atoms remaining = 367.88 atoms

Relative error = 0.9983%

Plot: Decay Curve with Half-Life and Time Constant Markers

```
ms      = 1.5;
t_cont = linspace(0, time(end), 500);
N_cont = N0 * exp(-t_cont / tau);

figure(1);
plot(time, N, '-ob', 'MarkerSize', ms); hold on;
plot(t_cont, N_cont, 'r--', 'LineWidth', 1.2);

xline(t_half_exp, '--g', 'LineWidth', 1.2);
yline(half_N, '--g', 'LineWidth', 1.2);
plot(t_half_exp, half_N, 'gs', 'MarkerSize', 8, 'LineWidth', 1.5);

xline(tau_exp, '--m', 'LineWidth', 1.2);
yline(target_N, '--m', 'LineWidth', 1.2);
plot(tau_exp, target_N, 'm^', 'MarkerSize', 8, 'LineWidth', 1.5);

xlabel('Time (s)');
ylabel('Number of Atoms');
title('Radioactive Decay — Euler Simulation');

legend('Euler (simulated)', 'Analytical N_0 e^{-t/\tau}', ...
      't_{1/2} marker', 'N_0/2 level', 'Half-life point', ...
      '\tau marker', 'N_0/e level', 'Time-constant point', ...
      'Location', 'northeast');

grid on;
hold off;
```

